

Israeli Financial Apps Reviews Analyzer

End-to-end data analytics pipeline that scrapes, enriches, and visualizes **47,188 Hebrew Google Play reviews** across **11 Israeli banking and credit card apps** over **15 years**. The work pairs T-SQL analysis on Microsoft SQL Server with Hebrew NLP, scikit-learn ML, and a three-page Power BI dashboard with a custom Navy and Gold theme. Portfolio analytics project. The full pipeline is reproducible from the bundled SQLite database and CSV exports without re-scraping or re-labeling.

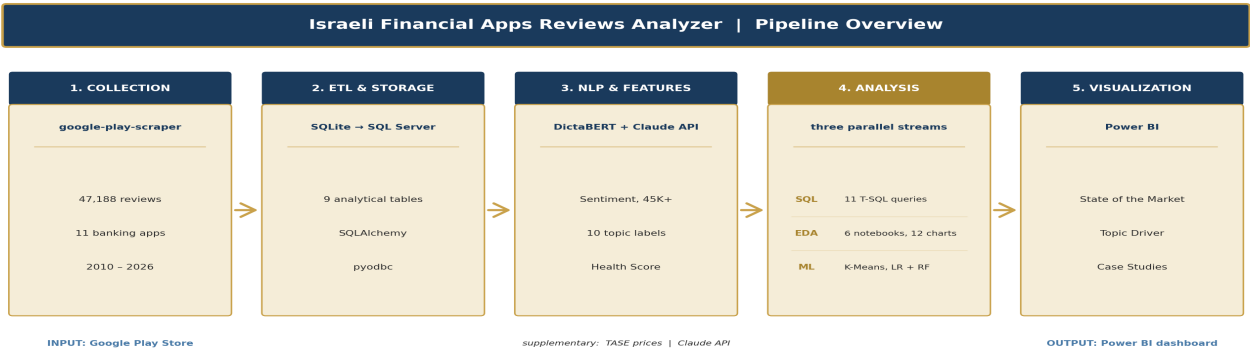


Figure 1. End-to-end pipeline from Google Play ingestion through SQL warehousing, Hebrew NLP enrichment, and the three-page Power BI dashboard.

The Business Question

Israeli banks compete on digital experience, yet leadership teams rarely see a unified, evidence-based view of how their app stacks up against the rest of the market. This project answers the operational question: **Which Israeli banking apps are winning on user experience, what drives complaints, and does that signal predict business outcomes?**

Four sub-questions an Israeli banking product, CX, or strategy team needs answered: **(1)** Who is winning? **(2)** Does the digital-native promise hold? **(3)** What drives complaints across 10 topic categories? **(4)** Does the app signal predict TASE stock returns?

Key Findings

- Digital banks beat traditional banks by 0.42 stars on average** (4.00 vs 3.58 stars). The gap is consistent year over year and shows up clearly in the per-app ranking.
- Reliability beats features as the lever that matters.** The four harshest topic categories are all foundational: Stability (83% negative), Login & Authentication (74%), Security (72%), Customer Service (72%). Feature complaints sit at 52%.
- Industry-wide decline since 2018.** Digital banks fell from 4.5 in 2021 to 2.0 in 2025. Pepper had the steepest individual fall: peak 4.50, trough 1.95.
- App quality is an operating KPI, not a stock signal.** Weekly Health Score versus weekly stock return: strongest correlation in the matrix is +0.12 (Isracard, contemporaneous), documented null finding across nine app-ticker pairs and the full -8 to +8 week lag sweep. Macro factors dominate at this horizon.

Bottom Line for a Stakeholder

The digital advantage is real, but fragile and shrinking. Digital-native banks lead by 0.42 stars on the segment average, yet Pepper itself fell from a 4.50 peak in 2021 to 1.95 in 2025. Incumbents that close the experience gap will compete on equal footing within three to five years.

Reliability beats features as the lever that matters. Engineering resources allocated to crash rates, auth flow reliability, and incident response will move the satisfaction needle more than any new feature. The complaint mix is the dashboard.

Treat app quality as an operating KPI, not a stock signal. Weekly correlations sit inside $|r| < 0.3$ across all nine app-ticker pairs and across the full -8 to +8 week lag sweep. App quality matters for retention, support cost, and brand. Treating it as a leading indicator for equity is overreach.

Methodology Highlights

Data layer. 47,188 reviews scraped from Google Play (IL, he) with resumable runs and JSON backups. Loaded into a nine-table SQLite warehouse with primary keys, foreign keys, and four indexes on the reviews table. Production warehouse on Microsoft SQL Server, migrated via ``import_to_sqlserver.py``.

NLP and feature engineering. DictaBERT classifies sentiment on 45,883 reviews. Anthropic Claude Haiku labels 31,109 reviews against 10 business categories using stratified sampling by sentiment. A composite Health Score per week per app combines average rating, sentiment balance, momentum, normalized volume, and developer response.

Analysis in three parallel streams. Eleven analytical T-SQL queries (CTEs, window functions, conditional aggregation) feed the Power BI semantic model. Six Jupyter notebooks tell the analytical story phase by phase. Three scikit-learn modeling layers: K-Means user segmentation with k auto-selected by silhouette score, score-change classifier (Logistic Regression vs Random Forest vs majority-class and regression-to-mean baselines), and a full lag sweep for stock correlation.

Skills Demonstrated

- **Advanced T-SQL.** Eleven analytical queries using CTEs, window functions, and conditional aggregation. Same queries drive both the EDA notebooks and the BI semantic model.
- **Power BI dashboard.** Three pages with custom Navy and Gold theme: State of the Market (KPIs, Health Score heatmap, peak-vs-current scatter), Topic Drivers, Case Studies (Leumi, Pepper, Isracard) with DAX measures and drill-through navigation.
- **Hebrew NLP at scale.** DictaBERT sentiment on 45,883 reviews. Claude API topic labeling across 10 business categories with stratified sampling.
- **ML modeling in scikit-learn.** K-Means user segmentation, score-change classifier, full -8 to +8 week lag sweep for stock correlation. Compared against principled baselines, not just defaults.
- **Production ETL.** SQLite working store, Microsoft SQL Server production warehouse, 9-table schema with PKs, FKs, indexed lookups, resumable scrapers, structured logging.
- **Analyst judgment.** Documented null finding on stock correlation. Explicit limitations on self-selection bias and the lack of stable user IDs. Findings framed as evidence rather than as marketing.

Tech Stack

Python 3.13. T-SQL on Microsoft SQL Server with CTEs, window functions, and conditional aggregation. SQLite via SQLAlchemy. pyodbc for SQL Server. Power BI Desktop with custom theme and DAX. transformers and torch for DictaBERT. Anthropic Claude API for topic labeling. scikit-learn for K-Means, Logistic Regression, and Random Forest. pandas, numpy, scipy for analytics. matplotlib, seaborn, and plotly for visualization. google-play-scraper and yfinance for ingestion.